

Query-Efficient Black-Box Adversarial Attack With Customized Iteration and Sampling

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Abstract—It is a challenging task to fool an image classifier based on deep neural networks under the black-box setting where the target model can only be queried. Among existing black-box attacks, transfer-based methods tend to overfit the substitute model on parameter settings. Decision-based methods have low query efficiency due to fixed sampling and greedy search strategy. To alleviate the above problems, we present a new framework for query-efficient black-box adversarial attack by bridging transfer-based and decision-based attacks. We reveal the relationship between current noise and variance of sampling, the monotonicity of noise compression, and the influence of transition function on the decision-based attack. Guided by the new framework, we propose a black-box adversarial attack named Customized Iteration and Sampling Attack (CISA). CISA estimates the distance from nearby decision boundary to set the stepsize, and uses a dual-direction iterative trajectory to find the intermediate adversarial example. Based on the intermediate adversarial example, CISA conducts customized sampling according to the noise sensitivity of each pixel to further compress noise, and relaxes the state transition function to achieve higher query efficiency. Extensive experiments demonstrate CISA's advantage in query efficiency of black-box adversarial attacks.

Index Terms—Adversarial perturbations, black-box attack, decision-based attack, transfer-based attack

1 INTRODUCTION

ADVERSARIAL examples [1], [2], [3] have revealed the vulnerability of deep neural networks (DNNs). Based on attackers' knowledge of the target model [4], adversarial attacks can be divided into white-box attacks and black-box attacks. In white-box attacks, the attackers can access detailed information of the target model, such as gradient information, training data, model structure, etc. However, in the more practical black-box setting [5], attackers can only query the target model and get the predictions without access to complete knowledge of the target model.

There are two inherent difficulties in black-box adversarial attacks. On the one hand, compared with white-box attacks, attackers under black-box setting cannot calculate gradients of the target model [4]. Therefore, it is less possible to use gradient-based or optimization-based attacks to directly find adversarial examples with small noise magnitude. On the other hand, the only way to obtain information from the black-box target model is to query, which is usually expensive and limited in number [5]. Under a limited

budget of query number to the target model, it is relatively difficult to compress the noise magnitude based on adversarial examples already found [6].

There are three types of black-box adversarial attacks: transfer-based attacks [7], [8], decision-based attacks [9], and attacks based on zeroth-order optimization [10], [11], [12], [13]. Among them, transfer-based attacks use adversarial examples generated on the local substitute model with white-box attack methods to attack the target model [14], [15], [16]. These methods exploit adversarial examples' transferability to circumvent the problem of not being able to calculate the gradient for the black-box target model. Decision-based attacks start from adversarial examples that have already been misclassified by the target model and gradually squeeze out noise by sampling in the input space of the original image [9], [17], [18]. Such attacks replace gradient calculation with sampling to find adversarial examples closer to the original image with query. Attacks based on zeroth-order optimization utilize gradient estimation to solve the problem that the target model cannot be back-propagated. This type of attack replaces the calculation of gradient for the target model by querying.

Considering the black-box attack setting where the query number is limited and the target model only returns hard label, existing attack methods have four serious drawbacks. *First, existing transfer-based attacks tend to overfit the substitute model* [8], [19]. Due to the differences in model structure and decision boundary between the substitute model and the target model, the adversarial examples generated using white-box attacks may only be effective on the substitute model. The attack parameters, such as the stepsize and iteration numbers, in a successful adversarial attack against the substitute model may not successfully misclassify the target model. *Second, existing transfer-based attacks have difficulties to*

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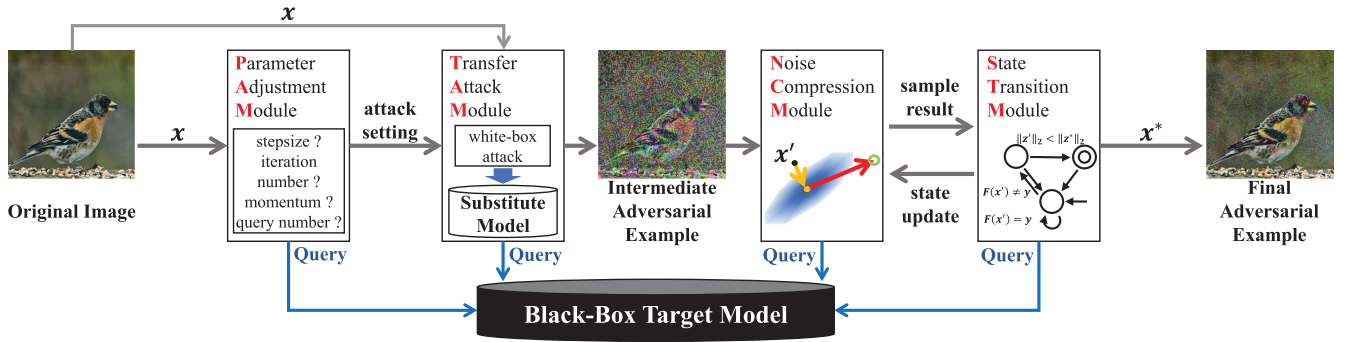


Fig. 1. The overview of our black-box adversarial attack framework. The attacker first uses PAM to adaptively set attack parameters to improve the query efficiency for transfer-based attacks. TAM generates intermediate adversarial examples on the substitute model according to parameters set by PAM. NCM takes the intermediate examples generated by transfer-based attack as the starting point for sampling, and uses the remaining number of queries with the assistance of STM to compress the magnitude of adversarial noise.

further compress the noises after generating adversarial examples. Many existing transfer-based attacks do not consider the query to the target model, and the generated adversarial examples are only related to the substitute model [14], [15], [16]. This hinders further optimization of adversarial examples and compression of adversarial noises based on the query results of the target model. Although there are methods [18], [20], [21] that combine transfer-based and decision-based attacks, there is still a lack of compression mechanism for redundant noise achieved by both transfer-based and decision-based attacks simultaneously. *Third, query efficiency of existing decision-based attacks is low.* Most existing decision-based attacks [18], [22] always sample on a constant distribution independent of historical queries or current noise, and never change the sampling during the attack process. This severely hinders the efficiency of noise compression. In addition, the stepsize of single-step modification in existing decision-based attacks is also a constant value. As the noise magnitude decreases, the success rate of the query will gradually decrease, and the efficiency of noise compression will be further affected with a constant stepsize. *Fourth, the attack process of existing decision-based attacks is prone to fall into local optimum.* In the process of querying and noise updating, existing decision-based attacks represented by Boundary Attack [9] only accept the adversarial examples with smaller noise magnitude as the starting point for new sampling. This greedy search process tends to fall into local optimum after several steps of compression, reducing the searching efficiency of later stages.

In this paper, we propose a query-efficient black-box adversarial attack framework which consists of four modules to cope with the above problems. As shown in Fig. 1, a Parameter Adjustment Module (PAM) and a Transfer Attack Module (TAM) are used to set the attack parameters and generate intermediate adversarial examples based on substitute model, respectively. After getting the intermediate adversarial examples, the Noise Compression Module (NCM) uses the remaining queries to search for adversarial examples with smaller noise magnitude, and adjusts the search direction with the help of State Transition Module (STM). The exploitation of gradient in transfer-based attacks and noise compression ability based on decision-based attacks are adapted under the setting of a limited query number. It is worth noting that a substitute model is not necessary for this framework. In the absence of a substitute

model, only the TAM needs to be replaced with other attacks that generate intermediate adversarial examples, such as gaussian noise. All existing transfer-based and decision-based attacks can be integrated within this general framework.

We further propose Customized Iteration and Sampling Attack (CISA) based on this new framework. First, we add Gaussian noise to the original image in the PAM. After finding an adversarial example close to the original image by continuously increasing the variance of Gaussian distribution, we set the stepsize according to the distance between this adversarial example and the original image. Second in the TAM, we iterate along both the gradient ascent and descent directions of the substitute model's loss function. The dual-direction setting curls up the iterative trajectories and is hence more likely to cross the target model's decision boundary at a closer distance. As for the NCM of the third step, we adaptively adjust the variance, mean, stepsize and mask of the sampling process for higher query efficiency. Instead of using unit variance for each dimension as in Boundary Attack, the variance of the multivariate normal distribution is linearly correlated to the absolute value of current noise. We customize the mean of sampling distribution by historical failed samples. In this way, new samples are guided away from directions with high failure rates. We also adjust the strategy of stepsize customization and mask of noise updating according to the pattern of noise compression. In addition, we relax the acceptance conditions of the decision-based attack in STM to reduce the probability that the searching process falls into a local optimum.

Beyond the new attack framework and CISA, we conduct a theoretical analysis of transfer-based attacks and decision-based attacks. We analyze the relationship between the sampling distribution of decision-based attack and noise compression, the monotonicity of noise compression, and the influence of transition function on convergence, which provides inspiration for the subsequent design of black-box adversarial attacks.

We perform comprehensive evaluations of existing adversarial attacks on the new framework and the effectiveness of CISA. Performance on ImageNet [23], Tiny-ImageNet [24] and CIFAR-10 [25] verifies that the new framework successfully improves the query efficiency of black-box attack with limited query number. Meanwhile, CISA generates adversarial examples with smaller perturbation in ℓ_2

norm than any combination of existing attack methods. Our code is available at <https://github.com/shiyuchengTJU/CISA>.

2 BACKGROUND

2.1 Notation

Consider a target model based on DNN under black-box attack: $F: X^N \rightarrow Y^C$, where X represents the input space, N is the dimension ($N = \text{Width} \times \text{Height} \times \text{Channel}$ for image data) and Y represents the classification space with C categories. A successful black-box adversarial attack minimizes the noise magnitude of adversarial example that misclassifies the target model under a limited number of queries

$$\min_{x' \in S_Q} \|x' - x\|_v, \quad \text{s.t. } F(x) \neq F(x') \text{ and } |S_Q| \leq T, \quad (1)$$

where x represents the original image, S_Q is the set of all adversarial examples to query the target model, T is the limit of query number. v refers to the norm used to measure the noise magnitude including ℓ_0 [26], ℓ_1 [27], ℓ_2 [9] and ℓ_∞ [7] norm. In this paper the ℓ_2 distance is calculated under the premise that adversarial examples are misclassified by the target model. The ℓ_2 norm is chosen as the distance metric because it characterizes the robustness of one model more accurately than ℓ_∞ norm [28].

The practicability of above black-box attack setting is two-fold. First, we assume the attacker has the ability to query the target model with limited number and get the output, which is more practical than a target model cannot be queried at all [15] or can be queried with a large number [10]. Second, we consider the case where the target model only outputs hard labels, which is more common than the case where the attacker can obtain the confidence or logits information of each category [4].

2.2 Transfer-Based Attack

Transfer-based attacks fool DNNs by exploiting transferability between local substitute model and the target model, i.e., phenomenon that adversarial examples generated by one model can fool another [5]. As a classical one-step attack, FGSM [7] finds the noise's direction by calculating the gradient of cross-entropy loss. I-FGSM [14] splits upper bound of noise into several small stepsize and adds noises step by step. MI-FGSM [15] introduced a momentum term to make the adjustment of the noise-adding direction smoother. Vr-IGSM [16] replaces the gradient of the original image with an averaged gradient of original image with gaussian noises. Gaussian noise eliminates local fluctuation in substitute model, thus improves the transferability. In addition to improving the transferability of black-box adversarial examples between models, there are also work that focuses on the universality of adversarial noises between images [29], [30].

2.3 Decision-Based Attack

Decision-based attacks sample in the neighborhood of the original image to seek smaller noise magnitude without crossing decision boundaries. Decision-based attacks do not rely on substitute models, but require an initial adversarial

example that has already been misclassified as starting point. Boundary Attack [9] starts from an gaussian noise and search along two directions simultaneously, namely source direction and spherical direction

$$x_{t+1} = x_t + \delta \cdot \frac{\eta}{\|\eta\|_2} + \varepsilon \cdot \frac{x - x_t}{\|x - x_t\|_2}, \quad \eta \sim \mathcal{N}(0, I), \quad (2)$$

where x_t is the adversarial example with smallest noise after t steps of Boundary Attack. η and $(x - x_t)$ refer to the direction of spherical and source direction, respectively. δ is the stepsize of spherical direction, and ε is the stepsize of source direction. Because of the indiscriminate use of standard normal distribution for each dimension, Boundary Attack cannot evaluate and exploit the differences of noise sensitivity between pixels.

Many decision-based attacks are developed on the basis of Boundary Attack. Among them, the Evolutionary Attack [17] reduces the dimension of sampling space by bilinear interpolation and restricting noise to the central part of images. Evolutionary attack performs better in tasks involving strong prior knowledge such as face recognition. Subspace optimization in QEBA [31] and gradient estimation in NonLinear-BA [32] are also applied to reduce the number of queries and achieve high query efficiency. AHA [33] gathers information from historical queries as the prior for current sampling. SurFree Attack [34] is based on the geometrical mechanism to get the biggest distortion decrease for a given direction to be explored under the assumption of a hyperplane boundary to achieve noise compression.

Some works optimize the sampling process of decision-based attacks from other perspectives. Whey optimization [35] divides adversarial noise into groups to reduce noise magnitude. The greedy searching process of Whey tends to fall into local optimum after several steps of compression, reducing searching efficiency of later stage. HopSkipJumpAttack [6] estimates the gradient direction using binary information at the decision boundary. As a variant of HopSkipJumpAttack, PopSkipJump [36] extends decision-based attack to probabilistic outputs. Tangent attack [37] is a geometric-based hard-label black-box attack that searches a boundary point along a tangent line of a virtual hemisphere. Sign-OPT attack [38] uses a zeroth order oracle to compute the sign of directional derivative of the attack objective using single query to achieve hard-label black-box attack.

2.4 Overlap of Transfer and Decision-Based Attacks

Recently, several methods involving both transfer-based and decision-based attack have emerged. LeBA [20] updates the substitute model of transfer-based attack with the confidence feedback of the decision-based attack. QAIR [21] uses decision-based attack to extract and approximate the target model for more transferable adversarial examples. DAIR [39] adopts an efficient sampling strategy based on substitute model to improve query efficiency against image retrieval models. The Biased Boundary Attack [18] uses the gradient from substitute model of transfer-based attack to concentrate samples of decision-based attacks on low-frequency domain of input space to make the adversarial example more

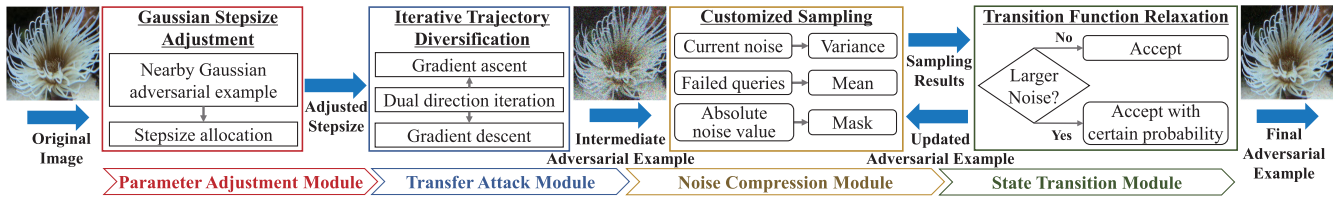


Fig. 2. The pipeline of CISA under the new black-box adversarial attack framework. CISA optimizes the parameter setting, the initialization of adversarial examples, the noise compression and the transition function according to PAM, TAM, NCM and STM, respectively.

‘natural’. These methods differ in goal from this paper. These combinations essentially use one black-box attack as an auxiliary or optimization of another, while the independent use of transfer-based and decision-based attacks in this paper is to find and compress adversarial noise with fewer queries.

3 OVERVIEW OF NEW BLACK-BOX ATTACK FRAMEWORK

In this section, we explain the function of the four modules in the framework and the overall pipeline.

3.1 Parameter Adjustment Module

As the first step of black-box attack framework, the Parameter Adjustment Module (PAM) adaptively sets attack parameters and strategies to alleviate the overfitting problem of transfer-based attacks to the substitute model. Since transfer-based attacks mainly rely on optimization or gradient-calculation on the substitute model to generate and update adversarial noises, the discrepancy on model structure and decision boundary burdens transferability between substitute model and target model [19], [40]. At the same time, there are usually multiple parameters in existing transfer-based attacks [14], [15], [16], such as the stepsize of iterative attack, the iteration number, etc. For transfer-based attacks, a set of parameters that successfully fool the substitute model may not misclassify the target model. We propose the Parameter Adjustment Module, which uses a combination of estimation and query to adapt the attack parameters for each original image exploiting the feedback from the target model. We will give an example of parameter adjustment method in Section 4.1.

3.2 Transfer Attack Module

The Transfer Attack Module (TAM) generates intermediate adversarial examples on the substitute model. TAM takes the original image as the starting point, and uses gradient-based or optimization-based attack methods to construct adversarial noises according to parameters given by PAM, and outputs adversarial examples that misclassify the target model. Almost all gradient-based [7], [14], [15], [16], [41], [42] or optimization-based attacks [43], [44], [45] can be embedded in TAM. In Section 4.2, we propose a new approach for TAM that diversifies the iterative trajectory to solve the problems of poor transferability and diminishing marginal effect of existing methods.

3.3 Noise Compression Module

The Noise Compression Module (NCM) further compresses the noise magnitude of the adversarial example. NCM takes the intermediate adversarial examples output by TAM as

the starting point, and searches for adversarial examples closer to original image by sampling in the neighborhood. Existing decision-based black-box attacks [6], [9], [17], [18], [35] are applicable to this module. In Section 4.3, we propose a new Customized Sampling scheme for NCM, which alleviates another problem of low query efficiency of existing decision-based attacks.

3.4 State Transition Module

The State Transition Module (STM) is used to adjust the update of the adversarial examples in the noise compression process. The decision-based attacks in NCM search in the neighborhood of an adversarial example, and uses the newly sampled adversarial example to query the target model. STM decides how to update the adversarial example for the next sampling of NCM based on whether the new adversarial example is misclassified by the target model and whether it is closer to the original image. The state transition strategy in the existing decision-based attacks [6], [9], [17], [18], [35] is relatively straightforward. If the new adversarial example fools the target model and is closer to the original image, it will replace the current adversarial example for new sampling. We will demonstrate in Section 4.4 that this greedy strategy may lead to the decision-based attack falling into a local optimum. We propose transition function customization for STM, using the relaxed state transition function to reduce the probability of decision-based attacks falling into local optimum.

Fig. 1 shows the pipeline of our new black-box adversarial attack framework. This black-box adversarial attack framework is general and open. The four modules are not limited to specific methods. Existing gradient-based and optimization-based attacks can all be embedded into TAM, and decision-based attacks can all be embedded into NCM. The framework is designed to provide a modular approach for combination, comparison and analysis for black-box attacks with limited query numbers. It is worth mentioning that this framework is not only for black-box adversarial attacks that include substitute models. When the substitute model is not available, NCM and STM still work properly by simply using some alternative noise, such as Gaussian noise, to generate the intermediate adversarial example.

4 CUSTOMIZED ITERATION AND SAMPLING ATTACK

In this section, we propose Customized Iteration and Sampling Attack (CISA) based on the new framework in Section 3. Fig. 2 demonstrates the general process of CISA. Given the original image, CISA first determines the stepsize with Gaussian Stepsize Adjustment, and uses the dual-direction iteration trajectory on the substitute model to

generate the first intermediate adversarial example that fools the target model. In the Customized Sampling, CISA compresses the noise magnitude of the intermediate adversarial example with the remaining query numbers. The variance, mean, mask and transition function of the sampling process are customized by the current noise, historical failed queries, number of successful queries, absolute value of current noise and remaining query numbers, respectively.

4.1 Gaussian Stepsize Adjustment

For parameter adjustment module, we propose Gaussian Stepsize Adjustment for the balance between stepsize and the iteration number in transfer-based attack. The stepsize μ is an important parameter in transfer-based attack. For an original image x , a large stepsize usually means fewer query times and large noise magnitude, while a small stepsize corresponds to smaller noise magnitude and more query times. There are only two straightforward strategies for stepsize setting in transfer-based attacks. The first is to set a fixed stepsize [14], [15], [16]. This strategy regards the stepsize as a hyperparameter of adversarial attacks. Take MI-FGSM [15] as example, the stepsize, number of iterations, and max noise magnitude are all manually set. Meanwhile, MI-FGSM uses the same attack parameter setting for all images in the test set. However, the minimum adversarial noise required to misclassify each image is not the same. Using the same setting may not generate adversarial examples with sufficiently small noise.

The second strategy is the binary search [46]. Specifically, the attacker first uses a small stepsize and iteratively attack the target model. If the generated adversarial example fails to misclassify the target model within a certain number of iterations, increase the stepsize and search in a larger input space. Once the attack is successful, the largest stepsize of unsuccessful iteration and smallest stepsize of successful iteration are used as two edge elements for binary search. The second strategy can find adversarial examples with smaller noise, but it consumes a large number of query numbers during multiple rounds of search. Such search strategy is inefficient and is not suitable for situations where multiple parameters are needed to be adjusted.

The essence of the stepsize setting problem of transfer-based attack lies on the estimation of the distance between original image and the nearby decision boundaries. First, because the shortest distances between different original images and decision boundaries are different on different target models, a fixed stepsize should not be an optimal choice. Second, for an iterative process, a balanced stepsize will not significantly increase the noise magnitude caused by the original image being too close to the decision boundary, nor will it waste too many queries if the decision boundary is far away.

Aiming at the drawbacks of these two existing stepsize setting strategies, we propose Gaussian Stepsize Adjustment, which neither uses a fixed stepsize for all images nor uses binary search to find the optimal stepsize. Before starting the iterative process of transfer-based attack, we try to add Gaussian noise to the original image to generate adversarial examples

$$x^{Gau} = Clip_{x,\tau}\{x + \xi^{Gau}\}, \quad \xi^{Gau} \sim \mathcal{N}(0, var^2 I), \quad (3)$$

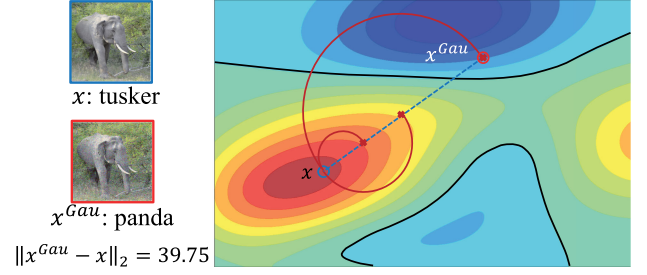


Fig. 3. Illustration of Gaussian Stepsize Adjustment.

where ξ^{Gau} refers to the random noise with the same dimension as the original image x and obeys the Gaussian distribution with mean of 0 and an initial variance of var_0 . I is an identity matrix of the same dimension as x . We initially set a small variance, and query the target model with the image x^{Gau} . If the output of the target model is not changed, we double the variance. Otherwise, an adversarial example based on Gaussian noise is found and we stop the search process. We refer to the ℓ_2 distance between x^{Gau} and x to set the stepsize for subsequent iteration

$$\begin{aligned} \hat{\mu} &= \min_{t \in \mathbb{N}} \|Clip_{x,\tau}\{x + \xi_t^{Gau}\} - x\|_2 / T, \\ \xi_t^{Gau} &\sim \mathcal{N}(0, (2^t \cdot var_0)^2 I), \\ s.t., F(x) &\neq F(Clip_{x,\tau}\{x + \xi^{Gau}\}), \end{aligned} \quad (4)$$

where $\hat{\mu}$ represents the stepsize obtained by Gaussian Stepsize Adjustment, and T refers to the iteration steps of the transfer-based attack. Since the noise magnitude of the adversarial example based on Gaussian noise is usually larger than that generated by iterative methods [47], setting $\|x^{Gau} - x\|_2 / T$ can basically ensure that the adversarial example can be found within T steps.

As shown in Fig. 3, CISA first uses Gaussian Stepsize Adjustment in PAM to find adversarial example based on gaussian noise x^{Gau} in the neighborhood of original image x , thereby setting the stepsize for TAM. The consecutive black curve represents decision boundary between original category 'tusker' and misclassified category 'panda'. The blue and red ring represent the original image x and Gaussian adversarial example used for stepsize adjustment. Gaussian Stepsize Adjustment adaptively allocates stepsize based on a nearby Gaussian adversarial example of each original image, which can effectively reduce the noise magnitude caused by a fixed iterative stepsize. In addition, since the variance is doubled once x^{Gau} is not misclassified by target model F , the adjusted stepsize $\hat{\mu}$ can be obtained with only a few queries. Compared with the binary search strategy, Gaussian Stepsize Adjustment significantly reduces the query number.

4.2 Iterative Trajectory Diversification

For Transfer Attack Module, we adapt existing adversarial attack methods to black-box settings with limited query numbers through diversified iterative trajectories. Two problems exist in gradient-based and optimization-based attacks. First of all, most attacks based on gradients or optimization do not consider huge gaps on classification spaces between different models. Their gradient directions may be

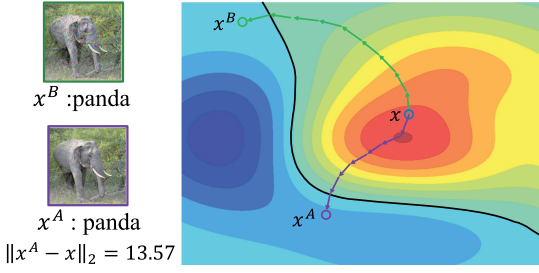


Fig. 4. Dual-direction iteration in Iterative Trajectory Diversification.

even orthogonal to each other [19]. Second, most transfer-based black-box attacks do not adapt to the setting with a limited query number. Taking iterative attack as an example, directly applying I-FGSM to black-box setting with limited query number may cause diminishing marginal effect. Assume that in order to minimize the noise magnitude, the stepsize μ of each step is inversely proportional to the total iteration numbers. In I-FGSM, when the query number T increases by 1, the marginal gain for the decrease in the noise magnitude is

$$\sum_{t=1}^{T+1} \frac{1}{T+1} \cdot \nabla J_{sub}(x_t) - \sum_{t=1}^T \frac{1}{T} \cdot \nabla J_{sub}(x_t). \quad (5)$$

In general, as T increases and the single stepsize shortens, the iterative trajectory tends to be consistent and smooth and gradually converges. Considering that the number of queries to the target model in black-box attack is also limited, increasing the iteration number has little effect on adversarial noise reducing if the iteration number is already high. In Section 4.2, we propose a new approach for TAM that diversifies the iterative trajectory to solve the problems of poor transferability and diminishing marginal effect of existing methods.

We attribute the diminishing marginal effect of transfer-based attack to the monotonicity of the iterative trajectory. First, monotonically employing gradient ascent along substitute model's loss function is more likely to bring iterative trajectories into local optimum of substitute model, rather than passing through the decision boundary of target model. Second, simply relying on transferability between substitute model and target model, but ignoring the feedback of target model after each query makes the iterative trajectories lack adaptability.

To diversify the iterative trajectory may be a more cost-effective solution [48]. Fig. 4 shows one possible distribution of target model loss function. In the case that loss function rises slowly along the direction of gradient ascent, like the green trajectory, it may be possible to find a shortcut across the decision boundary from a nearby starting point, as shown by the purple polyline in Fig. 4. We abandon the monotonic search strategy base on gradient ascent to increase the diversity of iterative trajectories

$$x'_0 = x, \quad x'_1 = \text{Clip}_{x,\tau}\{x'_0 - \mu \cdot \nabla J_{sub}(x'_0)\}, \quad (6)$$

$$g_{t+1} = \begin{cases} -\nabla J_{sub}(x'_t) & J_{sub}(x'_t) < J_{sub}(x'_{t-1}), \\ \nabla J_{sub}(x'_t) & J_{sub}(x'_t) \geq J_{sub}(x'_{t-1}), \end{cases} \quad (7)$$

$$x'_{t+1} = \text{Clip}_{x,\tau}\{x'_t + \mu \cdot g_{t+1}\}, \quad (8)$$

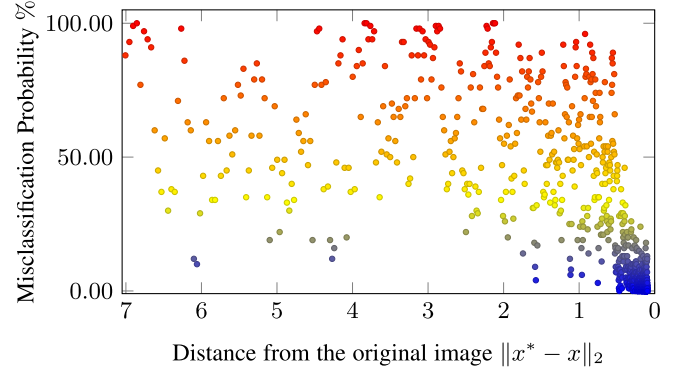


Fig. 5. The relationship between the distance from the original image and the misclassification probability.

where $J_{sub}(x'_t)$ represents the cross entropy loss of adversarial example x'_t on the substitute model. First, update the original image for one step along the direction of gradient descent. When the cross entropy loss of current adversarial example on substitute model is lower than the previous step, usually the local minimum of loss function has not yet been reached. Therefore, when the loss on the target model is still declining, continue to update along the direction of gradient descent, and vice versa.

In the actual implementation, in order to prevent the oscillation of adversarial noise update, we do not directly determine the gradient symbol on account of target model's loss function, but divide each iterative round into two stages. In the first stage, carry out gradient descent to the original image. Once the cross entropy on target model is lower than the previous step, the second stage starts and carries out gradient ascent until the last step. The normal iterative trajectory of direct gradient ascent is performed simultaneously. Inspired by vr-IGSM [16], we add gaussian noise to image in gradient calculation process to improve the transferability. Fig. 4 demonstrates the process of TAM. Background is contour of cross entropy loss. The redder the color, the lower the loss. Two polylines of green and purple represent iterative trajectories of gradient ascent and gradient descent, respectively. In TAM, CISA uses diversified iterative trajectories to add noise from two directions of gradient ascent and descent, and provides intermediate adversarial examples x' for NCM.

4.3 Customized Sampling

For Noise Compression Module, we customize the sampling process of decision-based attacks. First, we demonstrate the advantage of using adversarial examples from transfer-based attacks as the starting point of noise compression from the monotonicity of noise compression process for decision-based attacks.

We generate adversarial examples with different noise magnitudes for 1000 original images in ImageNet, and repeat Boundary Attack for each adversarial example 1000 times. The misclassification probability of Boundary Attack is reported in Fig. 5. It can be seen that in the neighborhood of original images, the probability of noise compression is positively correlated with the noise magnitude.

Under the assumption that misclassification probability increases monotonically with the distance from the original

image, the final noise magnitude is positively correlated with the initial noise magnitude. On the one hand, this explains the effectiveness of initializing adversarial noise by transfer-based attacks. On the other hand, we adapt the strategy of customize stepsize in Boundary Attack based on this feature.

According to the setting in [28], we denote $\rho_{F,x}(\lambda)$ as the misclassification probability of target model F for a random point with the distance λ from original image x

$$\rho_{F,x}(\lambda) = \mathbb{P}_{z \sim \lambda \mathbb{S}} \{F(x) \neq F(x+z)\}, \quad (9)$$

where $\lambda \mathbb{S}$ denotes the uniform measure on the sphere surface centered at 0 and of radius λ . In other words, the set of points satisfy $\|z\|_2 = \lambda$. Since the risk of misclassification generally increases with the distance from the original image, we assume $\rho_{F,x}(\lambda)$ (abbreviated as $\rho(\lambda)$) increases monotonically with λ within a certain range $\Delta_{adv}(x; F) \leq \lambda \leq \Delta_{unif,\kappa}(x; F)$

$$\begin{aligned} \forall \lambda_1, \lambda_2 \in [\Delta_{adv}(x; F), \Delta_{unif,\kappa}(x; F)], \\ \lambda_1 \geq \lambda_2 \rightarrow \rho(\lambda_1) \geq \rho(\lambda_2). \end{aligned}$$

As defined in [28], $\Delta_{adv}(x; F)$ denotes the ℓ_2 norm of the global smallest adversarial perturbation that causes misclassification. And $\Delta_{unif,\kappa}(x; F)$ denotes the κ -robustness of F to random uniform noise. For simplicity, we assume that after t steps of queries in decision-based attack, the noise magnitude of adversarial example x' to be queried is subject to a uniform distribution from $\Delta_{adv}(x; F)$ to the current smallest noise magnitude $\|z^*\|_2$.

Proposition 1. Assume that misclassification probability $\rho(\lambda)$ increases monotonically with λ within a certain range $\Delta_{adv}(x; F) \leq \lambda \leq \Delta_{unif,\kappa}(x; F)$. For any two adversarial examples x'_1 and x'_2 about original image x , if the corresponding noise magnitude satisfies $\Delta_{adv}(x; F) \leq \lambda_2 < \lambda_1 \leq \Delta_{unif,\kappa}(x; F)$, the expected noise magnitude after one step of decision-based attack satisfies $\mathbb{E}(\lambda_2) < \mathbb{E}(\lambda_1)$.

Proof 1. Consider the relationship between noise magnitudes $\lambda = \|z\|_2$ and misclassification probabilities $\rho(\lambda)$, the expected noise magnitude after $t+1$ steps is

$$\mathbb{E}(\lambda) = \int_{\Delta_{adv}(x; F)}^{\lambda} a \rho(a) \frac{1}{\lambda - \Delta_{adv}(x; F)} da. \quad (10)$$

Since it is impossible to get the specific value of $\Delta_{adv}(x; F)$ before the decision-based attack completely converges, and $\rho(\lambda) = 0$ when $\lambda \leq \Delta_{adv}(x; F)$, we rewrite Eqn. (10) as

$$\mathbb{E}(\lambda) = \int_0^{\lambda} a \rho(a) \frac{1}{\lambda} da. \quad (11)$$

For x'_1 and x'_2 satisfies $\Delta_{adv}(x; F) \leq \lambda_2 < \lambda_1 \leq \Delta_{unif,\kappa}(x; F)$, the difference between the expected noise magnitude after one step of decision-based attack is

$$\begin{aligned} \mathbb{E}(\lambda_1) - \mathbb{E}(\lambda_2) &= \int_0^{\lambda_1} a \rho(a) \frac{1}{\lambda_1} da - \int_0^{\lambda_2} a \rho(a) \frac{1}{\lambda_2} da \\ &= \frac{1}{\lambda_1} \int_{\lambda_2}^{\lambda_1} a \rho(a) da - \left(\frac{1}{\lambda_2} - \frac{1}{\lambda_1} \right) \int_0^{\lambda_2} a \rho(a) da. \end{aligned}$$

Given $\lambda_2 < \lambda_1$ and monotonicity of $\rho(\lambda)$, we have

$$\begin{aligned} \mathbb{E}(\lambda_1) - \mathbb{E}(\lambda_2) &\geq \frac{1}{\lambda_1} \int_{\lambda_2}^{\lambda_1} a \rho(a) da - \left(\frac{1}{\lambda_2} - \frac{1}{\lambda_1} \right) \int_0^{\lambda_2} a \rho(a) da \\ &= \frac{\rho(\lambda_2)}{2\lambda_1} (\lambda_1^2 - \lambda_2^2) - \frac{\lambda_2^2 \rho(\lambda_2)}{2} \left(\frac{1}{\lambda_2} - \frac{1}{\lambda_1} \right) \\ &= \frac{\rho(\lambda_2)}{2\lambda_1} (\lambda_1^2 - \lambda_1 \lambda_2) > 0. \end{aligned}$$

□

From Proposition 1, we show that under the assumption that misclassification probability increases monotonically with the distance from the original image, the expected noise magnitude after one step also increases monotonically with the initial noise magnitude. The decision-based attack process satisfies memorylessness, i.e., the current noise is determined only by noise of last step [9]. Therefore, the monotonicity of noise compression satisfies transitivity in multiple steps. In other words, when using the same decision-based attack and querying the target model for the same times, the expected final noise magnitude is positively correlated with the initial noise magnitude.

This explains the effectiveness of bridging transfer-based attack with decision-based attack in black-box scenario. Decision-based methods such as Boundary Attack use random noise as initial noise, whose magnitude is much larger than that generated by transfer-based attacks, thus the final noise is also larger under the same number of queries. Therefore, NCM uses adversarial examples generated by transfer-based attack as the starting point of decision-based attack. In Section 4.3, we propose a new Customized Sampling scheme for NCM, which alleviates another problem of low query efficiency of existing decision-based attacks.

Next, we demonstrate the insights for Customized Sampling by analyzing relationship between variance of multivariate normal distribution in Boundary Attack and the expectation of noise reduction. Suppose x^* is an adversarial example with smallest noise magnitude that we have found. The objective of decision-based attack can be described as

$$\max_x \|x^* - x\|_2 - \|x' - x\|_2, \quad s.t. \quad F(x) \neq F(x') \quad (12)$$

where x and x' represent the original image and the new adversarial example generated after this step, respectively. We replace the adversarial examples x^* and x' with the sum of original image x and adversarial noise, z^* and $z^* + z$, where z^* and z are the current adversarial noise with minimum magnitude and the noise added after this step, respectively. Since x and x^* are fixed, the objective function in Eqn. (12) can be equivalently reformulated as

$$\min_z \|z^* + z\|_2, \quad s.t. \quad F(x) \neq F(x + z^* + z) \quad (13)$$

Consider the one-step noise update of Boundary Attack in Eqn. (2), we rewrite z as $\delta \cdot \frac{\eta}{\|\eta\|_2} + \varepsilon \cdot \frac{x - x^*}{\|x - x^*\|_2}$, and $(1 - \frac{\varepsilon}{\|x - x^*\|_2})$ as α . Since $x - x^* = -z^*$, we have

$$\begin{aligned}
\|z^* + z\|_2 &= \left\| z^* + \delta \cdot \frac{\eta}{\|\eta\|_2} + \varepsilon \cdot \frac{-z^*}{\| -z^* \|_2} \right\|_2 \\
&= \left\| \alpha \cdot z^* + \delta \cdot \frac{\eta}{\|\eta\|_2} \right\|_2 \\
&= \sqrt{\|\alpha \cdot z^*\|_2^2 + 2 \cdot \delta \cdot \alpha \cdot (z^* \bullet \frac{\eta}{\|\eta\|_2}) + \|\delta \cdot \frac{\eta}{\|\eta\|_2}\|_2^2}
\end{aligned}$$

where $\bullet$ denotes the standard inner product. Since z^* , ε and δ are all fixed, $\|\delta \cdot \frac{\eta}{\|\eta\|_2}\|_2^2 \equiv \delta^2$, object of Eqn. (13) is actually to minimize $z^* \bullet \frac{\eta}{\|\eta\|_2}$. By the Cauchy-Schwarz Inequality we have

$$- \|z^*\|_2 \cdot \|\eta\|_2 \leq z^* \bullet \eta \leq \|z^*\|_2 \cdot \|\eta\|_2, \quad (14)$$

with $\beta = \frac{\eta}{\|\eta\|_2}$, this yields

$$\begin{aligned}
\|z^* + z\|_2^2 &\geq \|\alpha \cdot z^*\|_2^2 - 2 \cdot \alpha \cdot \delta \cdot (\|z^*\|_2 \bullet \|\beta\|_2) + \|\delta \cdot \beta\|_2^2 \\
&= (\|\alpha \cdot z^*\|_2 - \|\delta \cdot \beta\|_2)^2 \Rightarrow \|z^* + z\|_2 \\
&\geq \|\alpha \cdot z^*\|_2 - \|\delta \cdot \beta\|_2.
\end{aligned}$$

The equality holds when $z^* = -k\beta, k \in \mathbb{R}^+$. In other words, the magnitude of total noise $\|z^*\|_2$ after Boundary Attack is minimized when the direction of $\frac{\eta}{\|\eta\|_2}$ and current noise z^* are exactly reversed. Suppose that $\eta \sim \mathcal{N}(0, \Sigma)$ is a N -dimensional random vector that follows the normal distribution with zero mean and covariance matrix $\Sigma = \text{diag}(\sigma_1^2, \dots, \sigma_N^2)$ (in Boundary Attack, $\Sigma = I_N$). Each element of η is a univariate normal distribution, the mean is set to zero for better exploration in the sampling space [17]. Since $\beta = \frac{\eta}{\|\eta\|_2}$ and $\beta_i = \frac{\eta_i}{\sqrt{\eta_1^2 + \dots + \eta_N^2}}$. The ratio of β_i^2 's expectation satisfies

$$\begin{aligned}
E(\beta_1^2) : E(\beta_2^2) : \dots : E(\beta_N^2) &= E(\eta_1^2) : E(\eta_2^2) : \dots : E(\eta_N^2) \\
&= \text{Var}(\eta_1) : \text{Var}(\eta_2) : \dots : \text{Var}(\eta_N) = \sigma_1^2 : \sigma_2^2 : \dots \\
&: \sigma_N^2.
\end{aligned}$$

As one kind of rejection sampling [49], the Boundary Attack will only query the target model when noise reduction regard to one sampling is greater than zero, i.e., $z^* \bullet \eta \leq 0$. Therefore, the expectation of new noise x' after one-step Boundary Attack minimizes when $\sigma_i \propto |z_i^*|, 1 \leq i \leq N$.

To show the influence of σ on noise reduction more intuitively, we visualize the distribution of x' in two-dimensional space in Fig. 6. Blue vectors represent $x^* = (3, 1)$. Red marks indicate the distribution of x' after 1000 samplings under normal distribution with $\sigma_1 : \sigma_2 = 3 : 1$ (a) and $\sigma_1 : \sigma_2 = 1 : 1$ (b) centered on $(3, 1)$. The deeper the red, the denser samples are in the neighborhood. Black line charts at $x_1 = 0$ and $x_2 = 4$ are independent probability distribution P_{ind} of x_2 and x_1 , respectively. When the variance ratio of two dimensions $\sigma_1 : \sigma_2 = x_1^* : x_2^*$, x' concentrates in the opposite direction of x^* in (a). However, with an equal variance for each dimension in (b), x^* evenly distributes in all directions, which impedes the efficient compression of noise. This relationship suggests that, compared to standard normal distribution for all dimensions, sampling over normal distribution customized by current noise increases the expectation of noise reduction.

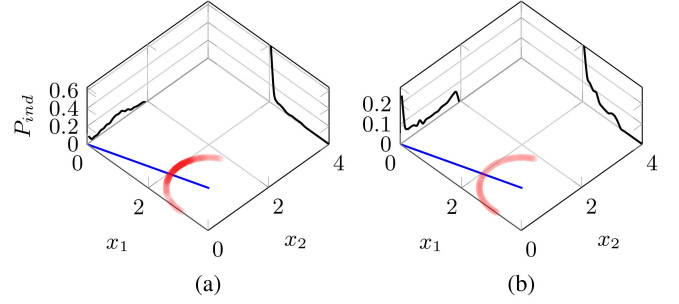


Fig. 6. Distribution of z with $\sigma_1 : \sigma_2 = 3 : 1$ (a) and $\sigma_1 : \sigma_2 = 1 : 1$ (b) when $N = 2$.

For Boundary Attack which randomly searches in a large input space, reducing the sampling space is critical to the efficiency of noise compression. Evolutionary Attack [17] reduces sampling space by bilinear interpolation and by limiting adversarial noise to the center of an image. Distinguishing the sensitivity of pixels to noise by relative position may be effective for images with a single structure (e.g., face recognition images) or a small size, but not for larger and more complex ones. The current noise z^* is a more unbiased characterization of the sensitivity of pixels compared to artificial rules in [17]. Therefore, we only adjust noise on pixels where the current noise magnitude is already large

$$H(z, r) = \arg \max_{\hat{z} \subset z^*, |\hat{z}|/|z^*|=r} \sum_{z \in \hat{z}} |z|, \quad (15)$$

$$mask_i = \begin{cases} 1, & \text{if } z_i^* \in H, \\ 0, & \text{else.} \end{cases} \quad (16)$$

where $\hat{z}$ is a set of pixels in z^* with the largest absolute values, $r \in (0, 1)$ is the ratio of the pixel numbers in $\hat{z}$ and z^* . Specifically, we pick pixels with the largest absolute value in z^* according to ratio r , and form up $mask$ to filter out the less sensitive area of the new noise.

Under the guidance of current noise, Customized Sampling adaptively assigns the variance ratio of the normal distribution for each dimension, and selects area most sensitive to noise to further reduce sampling space. Both processes take advantage of historical successful samplings. Although existing decision-based attacks directly discard failed samplings, they in fact contain information about decision boundaries. We amend the distribution's mean of next sample to keep away failed samplings

$$\begin{aligned}
\eta &\sim \mathcal{N}\left(-\frac{1}{K} \sum_{j=1}^K \tilde{\eta}_j, z^{*2}\right), \\
s.t. \quad F\left(x^* + \delta \cdot \frac{\tilde{\eta}}{\|\tilde{\eta}\|_2} + \varepsilon \cdot \frac{x - x^*}{\|x - x^*\|_2}\right) &= F(x),
\end{aligned}$$

where K is the total number of failed samplings on the current adversarial example x^* , and $\tilde{\eta}_j$ is the normal random vector used at the j th failed sampling. We maintain a sampling record of adversarial examples and save all the failed samplings about the current adversarial example x^* as $\tilde{x}$. The record is continually updated until a successful sampling occurs, i.e., the noise is further compressed. Since in the later stage of decision-based attack, the success rate of sampling decreases with the reduction of noise magnitude,

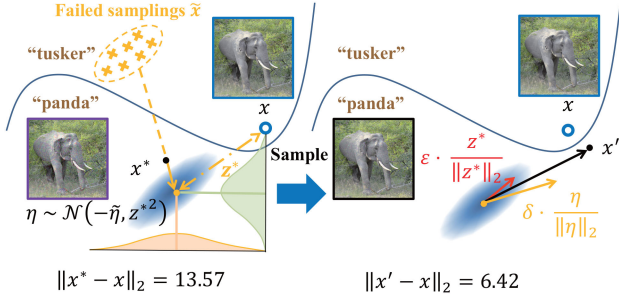


Fig. 7. Customization of variance in Customized Sampling.

maintaining the record can keep new samplings away from the historical failed ones, as demonstrated by yellow crosses in Fig. 7.

In addition, as the noise magnitude is continuously compressed, the possibility of misclassification for new query will gradually decrease if the stepsize δ and ε in Eqn. (2) for spherical and source direction remain the same. In order to compensate for the decrease in success rate of querying, we introduce the exponential scheduling to dynamically customize the stepsize in both directions

$$\delta_s = \delta_0 \varphi^s, \quad \varepsilon_s = \varepsilon_0 \varphi^s, \quad (17)$$

where s represents the number of successful queries so far, δ_s and ε_s are stepsizes of spherical and source direction after s successful queries. δ_0 and ε_0 are initial stepsizes. $\varphi \in (0, 1)$ is a decay factor for stepsize scheduling. As the distance between nearest adversarial example and the original image is shortened, the stepsize of new query is also reduced. This exponential scheduling strategy balances the noise compression rate with the query success rate, enabling stepsize customization for different images and for different query phases of one image.

Fig. 7 shows the customization of variance and mean in Customized Sampling. Blue curve represents decision boundary between 'tusker' and 'panda' categories. Yellow crosses are set of historical failed samplings $\tilde{x}$. Customized Sampling customizes the variance of the normal distribution by current noise z^* and its mean by opposite direction of historical failed samples $-\tilde{\eta}$. Then Customized Sampling performs random sampling on the customized distribution to form spherical direction. Finally, spherical (yellow arrow) and source directions (red arrow) are combined to generate an adversarial example x' with compressed noise magnitude.

4.4 Transition Function Relaxation

For State Transition Module, we relax the transition function of decision-based attacks to avoid search process from falling into local optimum. Taking the intermediate adversarial example x^* as the starting point, the set of all possible adversarial examples of Boundary Attack about the original image x and target model F can be denoted as

$$\begin{aligned} \mathbf{X}^P &= \{x^P \mid \exists x^o \in \mathbf{X}^P \cup \{x^*\}, \eta \sim \mathcal{N}(0, I) x^P \\ &= x^o + \delta \cdot \frac{\eta}{\|\eta\|_2} + \varepsilon \cdot \frac{x - x^o}{\|x - x^o\|_2}, s.t. F(x) \neq F(x^P)\}. \end{aligned}$$

For $x^o \in \mathbf{X}^P \cup \{x^*\}$, rewrite $\tilde{x} = x^o + \delta \cdot \frac{\eta}{\|\eta\|_2} + \varepsilon \cdot \frac{x - x^o}{\|x - x^o\|_2}$. The transition function of Boundary Attack on x^o at step t can be expressed as

$$B_t(x^o) = \begin{cases} \tilde{x}, & \text{if } F(x) \neq F(\tilde{x}) \text{ and} \\ & \|x - \tilde{x}\|_2 \leq \|x - x^o\|_2, \\ x^o, & \text{else,} \end{cases} \quad (18)$$

The noise update process of Boundary Attack has nothing to do with the attack trajectory. It is only related to the current state. Denote $V(t)$ as the adversarial example in Boundary Attack that have been updated for $t \geq 0$ steps, $V(0) = x$

$$\begin{aligned} Pr(B_{t+1}(x_t) = x^P \mid V(1) = x_1, V(2) = x_2, \dots, V(t) = x_t) \\ = Pr(B_{t+1}(x_t) = x^P \mid V(t) = x_t). \end{aligned}$$

where $0 \leq \omega \leq 1$ is the probability of accepting the update when getting an adversarial example with bigger noise magnitude, and $random()$ is a random number generator over $[0, 1]$. In the following proposition, we analyze the advantage of adversarial noise convergence when using the relaxed transition function of Eqn. (22) compared with the existing decision-based attacks based on greedy strategy that only accepts an adversarial example with smaller noise magnitude:

Proposition 2. Assume that the source stepsize is smaller than spherical stepsize $\varepsilon < \delta$. If the update on adversarial example x^o is accepted with the probability of $0 < \omega \leq 1$ when the updated adversarial noise is larger than that of x^o , i.e., $\|x - \tilde{x}\|_2 \geq \|x - x^o\|_2$, then $\lim_{m \rightarrow \infty} U_{i,j}^m(t) = \pi_j(t)$ exists for all $x_i, x_j \in \mathbf{X}^P$.

Proof 2. According to the definition of set $\mathbf{X}^P$, each element $x^P \in \mathbf{X}^P$ can be generated by updating another element in $\mathbf{X}^P$ through a single-step Boundary Attack. And because $\varepsilon \leq \delta$, $\eta \sim \mathcal{N}(0, I)$, the transition probability between any two adversarial examples $x_i, x_j \in \mathbf{X}^P$ exists when step $t = 0$

$$\forall x_i, x_j \in \mathbf{X}^P, \exists m \in \mathbb{N}, U_{i,j}^m(0) > 0. \quad (19)$$

When $t > 0$, since the relaxed transition function accept both the increase and decrease of the noise magnitude, the set of all possible adversarial examples remain unchanged. Therefore, for every finite step t , the transition function may assign a path of nonzero probability between any two adversarial examples $x_i, x_j \in \mathbf{X}^P$. The transition function satisfies irreducibility.

For any adversarial example $x_i \in \mathbf{X}^P$, because $\varepsilon < \delta$

$$\forall m \geq 2, \pi_i(t) > 0 \rightarrow U_{i,i}^m(t) > 0. \quad (20)$$

The greatest common divisor (gcd) of period for any state satisfies

$$\forall x_i \in \mathbf{X}^P, \gcd\{m \geq 1 : U_{i,i}^m(t) > 0\} = 1. \quad (21)$$

Therefore, the transition function satisfies aperiodicity. Because the update on adversarial example of Boundary Attack with the relaxed transition function Eqn. (22) is a irreducible and aperiodic Markov chain with finitely many states ($|\mathbf{X}^P| \ll 256^N$, where N is the dimension of input space), $\lim_{m \rightarrow \infty} U_{i,j}^m(t) = \pi_j(t)$ exists for all $x_i, x_j \in \mathbf{X}^P$ according to Theorem 2.1 in [50]. $\square$

Algorithm 1. Customized Iteration and Sampling Attack

Input: Target model $F(x)$, substitute model $Sub(x)$
 Original image x and label y
 Max querying number T and noise magnitude limit τ
 Variance of gaussian noise in gradient calculation R
 Initial Gaussian variance var for stepsize adjustment
 Identity matrix I of the same dimension as x
 Initial spherical stepsize δ_0 and source stepsize ε_0
 Decay factor for sampling φ and pixel retention rate r
 Random number generator $random()$ over $[0, 1]$

Output: Adversarial example x^* with compressed noise

// PAM: Gaussian Stepsize Adjustment

while $F(x^{Gau}) = y$ **do**
 $\xi^{Gau} \sim \mathcal{N}(0, var^2 I)$, $x^{Gau} \leftarrow Clip_{x,\tau}\{x + \xi_0^{Gau}\}$
 $var \leftarrow var * 2$, $T \leftarrow T - 1$
 $\hat{\mu} \leftarrow \|x^{Gau} - x\|_2 / T$, $x_0^A \leftarrow x$, $x_0^B \leftarrow x$, $downhill \leftarrow True$

// TAM: Iterative Trajectory Diversification;

while $T > 0$ **do**
 $\xi_t^A, \xi_t^B \sim \mathcal{N}(0, R^2 I)$
 $g_t^A \leftarrow \nabla J_{sub}(x_t^A + \xi_t^A)$, $g_t^B \leftarrow \nabla J_{sub}(x_t^B + \xi_t^B)$
 $x_{t+1}^A \leftarrow \begin{cases} Clip_{x,\tau}\{x_t^A - \hat{\mu} \cdot g_t^A\} & downhill = True \\ Clip_{x,\tau}\{x_t^A + \hat{\mu} \cdot g_t^A\} & downhill \neq True \end{cases}$
 $x_{t+1}^B \leftarrow Clip_{x,\tau}\{x_t^B + \hat{\mu} \cdot g_t^B\}$
if $J_{sub}(x_{t+1}^A) > J_{sub}(x_t^A)$ **then**
 $downhill \leftarrow False$

if $F(x_t^A) \neq y$ **then**
 $x' \leftarrow x_t^A$, $T \leftarrow T - 1$, **Break**

if $F(x_t^B) \neq y$ **then**
 $x' \leftarrow x_t^B$, $T \leftarrow T - 1$, **Break**

$T \leftarrow T - 2$

// NCM: Customized Sampling

$W \leftarrow []$, $s \leftarrow 0$, $x^* \leftarrow x'$, $z^* \leftarrow x^* - x$

while $T > 0$ **do**
if $W \neq \emptyset$ **then**
 $\eta \sim \mathcal{N}(-\frac{1}{|W|} \sum \tilde{\eta}, z^{*2})$ s.t. $\tilde{\eta} \in W$
else
 $\eta \sim \mathcal{N}(0, z^{*2})$

// Pick up pixels with the largest absolute value in z^*
 $H(z, r) = \arg \max_{\tilde{z} \subset z^*, |\tilde{z}|/|z^*|=r} \sum_{z \in \tilde{z}} |z|$
 Construct $mask$ by H according to Eqn. (15)
 $x' = x^* + mask \cdot (\delta_s \cdot \frac{\eta}{\|\eta\|_2} - \varepsilon_s \cdot \frac{z^*}{\|z^*\|_2})$, $T \leftarrow T - 1$

if $F(x') \neq y$ **then**
 // STM: Transition Function Customization
if $\|x' - x\|_2 < \|x^* - x\|_2$ OR $random() \leq \frac{t}{T}$ **then**
 $x^* \leftarrow x'$, $W \leftarrow []$, $s \leftarrow s + 1$, $\delta_s \leftarrow \delta_{s-1}\varphi$, $\varepsilon_s \leftarrow \varepsilon_{s-1}\varphi$

else
 $W \leftarrow W \cup \eta$

return x^* .

$$B_t(x^o) = \begin{cases} \tilde{x}, & \text{if } F(x) \neq F(\tilde{x}) \text{ and} \\ & \|x - \tilde{x}\|_2 \leq \|x - x^o\|_2, \\ \tilde{x}, & \text{if } F(x) \neq F(\tilde{x}) \text{ and} \\ & \|x - \tilde{x}\|_2 > \|x - x^o\|_2, \text{ and} \\ & random() \leq \omega, \\ x^o, & \text{else,} \end{cases} \quad (22)$$

As the Boundary Attack progresses, the scale of all possible adversarial examples of Boundary Attack is decreasing because the transition function Eqn. (18) without relaxation only accepts adversarial examples with smaller noise magnitude. In this case, if current adversarial example x_j at step t has the smallest noise among adversarial examples can be reached in one step, then an adversarial example x_i with smaller noise in the entire adversarial example space is not reachable

$$\begin{aligned} & \forall x_k \in \mathbf{X}^P, U_{k,j}(t) > 0 \rightarrow \|x - x_j\|_2 < \|x - x_k\|_2 \\ \bigwedge & \exists x_i \in \mathbf{X}^P, \|x - x_i\|_2 < \|x - x_j\|_2 \\ \Rightarrow & \forall m \in \mathbb{N}, U_{i,j}^m(t) = 0. \end{aligned}$$

From Proposition 2, we show that under the relaxed transition function of Eqn. (22), it is impossible for the attack process to fall into such local optimum. Because any two adversarial examples $x_i, x_j \in \mathbf{X}^P$ are reachable during the entire attack process. Therefore, we use the relaxed transition function of Eqn. (22) in Customized Sampling, whose smallest adversarial noise that may be found is smaller than or equal to the smallest noise under the original transition function Eqn. (18). Since the success rate of sampling decreases with the reduction of noise magnitude, it is easier to fall into local optimum in the later stage of decision-based attacks. Therefore, we set $\omega = t/T$ in the process of Customized Sampling, where t and T are the current query number and total query number. In other words, the probability of accepting adversarial examples with larger noise magnitude gradually increases as the sampling process proceeds to reduce the probability of decision-based attack falling into local optimum. CISA relaxes the transition function to accept adversarial examples of larger noise magnitude with a certain probability, and updates the sampling starting point of NCM. Algorithm 1 details CISA.

5 EXPERIMENTS

5.1 Experiment Settings

We conduct experiments on three image classification datasets: ImageNet [23], Tiny-Imagenet [24], and CIFAR-10 [25]. The image shapes of three datasets are $224 \times 224 \times 3$, $64 \times 64 \times 3$ and $32 \times 32 \times 3$, respectively. ImageNet contains 1000 image categories. We pick 10000 images from its validation set that can be correctly classified by all target models, 10 images for each category. As for Tiny-Imagenet with 200 image categories, we choose 2000 images, 10 images for each category. All 10000 images in the test set on CIFAR-10 are tested. 8 neural network models with different structures are compared: resnet-18 [51], resnet-101, inception v3 [52], inception-resnet v2 [53], nasnet [54], densenet-161 [55], vgg19-bn [56], senet-154 [57].

We compare 7 existing transfer-based attacks and 8 decision-based attacks with CISA. Among them, transfer-based

Therefore, the noise update process of Boundary Attack satisfies the Markov property. Let $U_{i,j}(t)$ be the probability of going from adversarial example x_i to x_j in one step at step t , $x_i, x_j \in \mathbf{X}^P$. And let $U_{i,j}^m(t)$ be the probability of going from adversarial example x_i to x_j in m steps at step t . Let $\pi_j(t) = U_{0,j}^t(t)$ be the probability that the adversarial example is exactly x_j after t steps of update, we have:

$$\forall x_j \in \mathbf{X}^P, \pi_j(t+1) = \sum_{x_i \in \mathbf{X}^P} \pi_i(t) U_{i,j}(t), \sum_{x_j \in \mathbf{X}^P} \pi_j(t) = 1.$$

If the condition for accepting the current noise update in Eqn. (18) is relaxed:

TABLE 1
Comparison Under the New Black-Box Adversarial Attack Framework on Tiny-Imagenet With res-18 as Target and inc-V3 as Substitute

			Target Model: res-18						Substitute Model: inc-v3					
			STM	N/A						NCM				Relax
					Vanilla	Boundary	Whey	BBA	EVO	HSJA	Tangent	Sign-OPT	QEBA	
PAM														
TAM	N/A	Random	Mid	1.663	0.463	0.677	0.294	0.432	0.856	0.637	1.185	0.470	0.224	
			Avg	3.867	1.128	1.603	0.986	1.134	2.686	2.344	3.278	2.126	0.903	
		DDN	Mid	0.739	0.431	0.408	0.317	0.439	0.587	0.488	0.592	0.398	0.276	
			Avg	2.407	1.174	1.29	1.068	1.173	2.029	1.905	2.276	1.668	0.949	
		FGSM	Mid	2.996	1.181	1.151	1.241	1.17	1.975	1.052	1.564	0.952	0.701	
			Avg	4.441	1.805	2.058	1.879	1.814	3.073	3.306	4.259	2.939	1.225	
		I-FGSM	Mid	0.698	0.443	0.397	0.308	0.425	0.548	0.552	0.604	0.508	0.285	
			Avg	2.09	1.568	1.705	1.529	1.563	2.05	1.460	1.635	1.395	1.346	
	Binary Search	MI-FGSM	Mid	0.783	0.532	0.451	0.38	0.508	0.655	0.556	0.748	0.488	0.341	
			Avg	2.243	1.298	1.401	1.312	1.313	1.958	1.730	2.064	1.559	1.072	
		vr-IGSM	Mid	0.594	0.348	0.316	0.243	0.358	0.5	0.435	0.541	0.376	0.187	
			Avg	1.861	0.924	1.092	0.807	0.921	1.827	1.439	1.708	1.294	0.805	
		C&W	Mid	0.432	0.35	0.35	0.261	0.387	0.532	0.385	0.432	0.363	0.231	
			Avg	22.697	30.609	30.622	30.583	30.61	30.654	31.772	31.793	31.766	30.58	
		EAD	Mid	0.524	0.383	0.346	0.237	0.345	0.477	0.324	0.478	0.294	0.237	
			Avg	1.613	1.025	0.914	0.791	1.016	1.39	1.199	1.475	1.099	0.707	
	GSA	CI	Mid	0.898	0.341	0.298	0.219	0.336	0.459	0.289	0.353	0.276	0.186	
			Avg	2.465	0.91	0.896	0.706	0.912	1.571	0.953	1.135	0.887	0.706	

attacks include FGSM [7], I-FGSM [14], MI-FGSM [15], vr-IGSM [16], DDN [41], C&W [43], and EAD [45]. Decision-based attacks include Boundary Attack [9], Whey [35], Biased Boundary Attack (BBA) [18], Evolutionary Attack (Evo) [17], HSJA [6], Tangent Attack [37], Sign-OPT Attack [38], and QEBA [31]. For hyperparameters, the stepsizes are adjusted by binary search for FGSM, I-FGSM, MI-FGSM, and vr-IGSM. Stepsizes of spherical direction and source direction are $\delta_0 = 0.1$, $\varepsilon_0 = 0.003$ for Boundary, BBA, Evo and CISA. Decay factor φ is set to 0.99. For BBA [18], we use the version that incorporates information from a substitute model at each step. Pixel retention rate $r = 0.2$ for our CISA. The initial Gaussian variance var Iterative Trajectory Diversification is 1. Variance in gradient calculation R is 1. It is worth mentioning that, since there is no similar black-box attack framework yet, we compare CISA with the combination of existing black-box transfer-based and decision-based attacks to verify our new framework. Since these transfer-based and decision-based attacks are not designed for this framework like CISA, the experimental results can only represent the performance under this framework, and cannot directly reflect the performance between stand-alone attack methods.

For evaluation criterion, we choose the median and average size of adversarial perturbation, as applied in NIPS 2018 Adversarial Vision Challenge [24]

$$mid = median(\{\|x' - x\|_2 \mid x \in \mathbf{X}\}), \quad (23)$$

$$avg = \frac{1}{n_{data}} \sum_{i=1}^{n_{data}} (\{\|x'_i - x_i\|_2 \mid x \in \mathbf{X}\}), \quad (24)$$

where n_{data} is the number of images in a dataset, x is an original image in the dataset $\mathbf{X}$. x' is the adversarial example found that is closest to x . A smaller ℓ_2 distance indicates a

better adversarial example. It is worth noting that adversarial examples are rounded before being input to the target model for a more realistic black-box attack setting.

5.2 Comparison Under New Framework

We embed existing transfer-based and decision-based attacks into the corresponding modules of the new black-box adversarial attack framework to benchmark the black-box attack abilities of different methods under limited query numbers. We report the median and average adversarial perturbation on Tiny-Imagenet in Tables 1 and 2, on ImageNet in Tables 3 and 4 and CIFAR-10 on Table 5. PAM is divided into three cases: 'N/A' means that the parameter adjustment module is not included, 'Binary Search' means the stepsize that minimizes the noise magnitude is found through binary search, and the 'GSA' represents Gaussian Stepsize Adjustment proposed in Section 4.1. We include 9 cases in total for TAM. In addition to the 7 existing transfer-based attacks, 'Random' represents the case where transfer-based attacks are not used and Gaussian noises are added to original images until misclassified by the target model. We denote the process before CISA generates intermediate adversarial examples as 'CI,' and process after generating intermediate adversarial examples as 'CS'. CI includes Gaussian Stepsize Adjustment and iteration trajectory diversification. CS includes the Customized Sampling and Transition Function Relaxation. C&W is not compared in ImageNet because its adversarial examples can hardly transfer to the target model successfully on a complex dataset. Apart from 8 existing decision-based attacks, NCM includes the Customized Sampling (CS) we proposed, and the case that directly outputs adversarial examples generated by transfer-based attacks without noise compression (Vanilla). Finally, STM include 3 cases. 'N/A' means that

TABLE 2
Comparison Under the New Black-Box Adversarial Attack Framework on Tiny-Imagenet With Inc-Res as Target and Nasnet as Substitute

			Target Model: inc-res						Substitute Model: nasnet				
			STM	N/A	NCM								Relax
					Greedy Search								
PAM			Vanilla	Boundary	Whey	BBA	EVO	HSJA	Tangent	Sign-OPT	QEBA	CS	
TAM	N/A	Random	Mid	6.42	1.474	2.035	1.209	1.389	3.444	1.502	1.922	1.340	0.944
			Avg	6.9	2.303	2.87	2.048	2.271	4.78	2.350	3.274	2.306	1.998
		DDN	Mid	1.407	1.148	1.009	0.983	1.165	1.34	1.243	1.311	1.147	0.75
			Avg	3.971	2.284	2.631	2.074	2.266	3.123	2.207	2.673	2.024	1.985
		FGSM	Mid	3.893	1.755	2.03	1.895	1.767	3.538	1.944	2.030	1.585	1.106
			Avg	8.014	3.218	4.289	3.258	3.294	5.307	2.955	3.607	2.714	2.685
		I-FGSM	Mid	1.947	1.527	1.369	1.189	1.437	1.79	1.444	1.652	1.266	0.974
			Avg	3.942	3.988	3.888	3.874	4.028	4.667	3.891	4.163	3.745	3.542
	Binary Search	MI-FGSM	Mid	2.245	1.494	1.388	1.376	1.437	2.048	1.613	1.896	1.541	1.01
			Avg	5.095	3.169	3.806	3.34	3.102	4.177	3.159	3.718	2.868	2.741
		vr-IGSM	Mid	1.235	1.131	0.832	0.785	1	1.163	0.968	1.112	0.866	0.61
			Avg	3.488	1.814	1.737	1.603	1.855	2.718	2.066	2.412	1.856	1.489
		C&W	Mid	1.375	1.912	1.842	1.571	1.888	2.144	1.809	1.854	1.525	1.483
			Avg	28.329	39.867	39.863	39.831	39.864	39.899	38.624	38.662	38.601	39.812
		EAD	Mid	1.116	0.988	0.838	0.81	1.145	1.373	0.922	1.100	0.827	0.575
			Avg	2.055	1.799	1.604	1.502	1.791	2.529	1.624	1.877	1.494	1.307
	GSA	CI	Mid	1.705	0.974	0.767	0.75	0.983	1.091	0.772	0.792	0.691	0.568
			Avg	3.07	1.436	1.259	1.315	1.418	1.789	1.529	1.662	1.457	1.057

the state transition is not available because noise compression is not performed. ‘Greedy Search’ means that the sampling starting point is updated only when the target model is misclassified with a smaller noise magnitude. ‘Relax’ represents the Transition Function Relaxation strategy we proposed.

For each in Tables 1, 2, 3, 4 and 5, the first row represents the target model and substitute model used. Each row below represents a transfer-based attack and its associated PAM, and each column represents a noise compression

method and its associated state transition strategy. For fair comparison, the noise magnitudes reported for all methods are the results after 1000 queries on the target model for each image. Except for ‘Random’ and ‘CI,’ other transfer-based attacks perform 1000 queries on the target model in the ‘Vanilla’ case, and 500 queries in other cases to leave the remaining 500 queries for decision-based attack in the NCM. Specifically, we try 500/1000 different stepsizes for FGSM and choose the smallest noise. For I-FGSM, MI-FGSM, vr-IGSM, C&W, and EAD we use a binary search

TABLE 3
Comparison Under the New Black-Box Adversarial Attack Framework on ImageNet With inc-101 as Target and vgg-19 as Substitute

			Target Model: res-101						Substitute Model: vgg-19				
					NCM								
			STM	N/A	Greedy Search							Relax	
PAM			Vanilla	Boundary	Whey	BBA	EVO	HSJA	Tangent	Sign-OPT	QEBA	CS	
TAM	N/A	Random	Mid	6.42	42.31	15.011	25.451	11.748	15.308	32.990	25.649	26.954	20.1503
			Avg	6.9	42.634	17.236	27.828	17.109	17.223	32.275	21.340	25.492	19.0403
	DDN	Mid	1.407	7.958	6.682	6.583	4.824	6.671	7.719	6.899	7.537	6.2715	
		Avg	3.971	13.09	9.837	10.891	8.194	9.925	12.881	9.156	11.240	8.70178	
	FGSM	Mid	3.893	13.334	8.048	8.676	6.501	7.841	11.366	8.836	10.439	7.92178	
		Avg	8.014	20.569	12.429	16.382	9.779	12.304	17.907	10.325	10.333	10.3218	
	I-FGSM	Mid	1.947	5.889	5.302	5.152	4.108	5.286	5.616	4.746	5.117	4.21308	
		Avg	3.942	10.839	8.334	8.959	6.887	8.322	9.510	8.655	10.534	7.4992	
	Binary Search	MI-FGSM	Mid	2.245	6.035	5.079	4.764	4.33	5.095	5.942	5.021	5.038	3.81972
			Avg	5.095	11.499	9.007	10.107	7.837	9.032	10.914	7.648	10.655	6.5637
		vr-IGSM	Mid	1.235	5.833	5.082	4.897	3.896	5.252	5.764	5.440	5.606	3.10856
			Avg	3.488	10.258	7.7	7.43	5.634	7.716	9.572	8.181	9.465	7.08316
		EAD	Mid	1.116	4.763	5.904	6.046	3.869	5.887	6.909	3.830	3.977	3.928
			Avg	2.055	8.706	9.048	9.294	6.716	9.021	11.219	6.181	8.254	5.28168
	GSA	CI	Mid	1.705	7.976	4.189	4.05	3.191	4.166	4.267	5.681	6.069	5.1892
			Avg	3.07	12.855	6.817	7.051	5.485	6.819	8.038	6.438	7.664	5.6016

TABLE 4
Comparison Under the New Black-Box Adversarial Attack Framework on ImageNet With Densenet as Target and Senet as Substitute

			Target Model: densenet						Substitute Model: senet				
			STM	N/A	NCM						Relax		
					Vanilla	Greedy Search							
PAM					Boundary	Whey	BBA	EVO	HSJA	Tangent	Sign-OPT	QEBA	CS
TAM	N/A	Random	Mid	46.249	15.818	25.693	13.124	16.108	33.276	21.480	22.560	17.640	10.971
			Avg	46.21	19.041	31.598	19.448	19.467	35.224	22.703	21.160	18.370	13.551
		DDN	Mid	13.047	9.566	9.642	6.89	9.029	12.388	10.100	12.863	8.646	6.574
			Avg	20.32	12.281	15.659	10.299	12.327	18.438	15.318	18.762	13.925	10.138
		FGSM	Mid	27.867	14.538	21.602	12.164	14.875	24.31	14.202	14.866	12.441	10.324
			Avg	34.746	18.818	27.817	17.501	19.19	29.038	24.101	24.858	22.173	15.244
		I-FGSM	Mid	6.68	5.551	5.559	4.734	5.568	5.962	5.023	6.532	4.442	4.208
			Avg	9.866	8.263	8.178	7.123	8.277	9.102	7.833	9.420	7.760	6.948
	Binary Search	MI-FGSM	Mid	6.091	5.353	4.991	4.369	5.388	5.72	5.213	5.583	4.320	4.24
			Avg	10.718	8.596	9.453	7.67	8.641	10.19	8.611	8.702	8.448	7.257
		vr-IGSM	Mid	5.643	4.71	4.503	3.518	4.76	4.918	4.988	5.510	4.623	3.208
			Avg	9.675	7.442	8.22	6.086	7.51	8.795	7.404	8.728	6.561	5.827
		EAD	Mid	5.459	4.721	4.922	3.839	4.811	4.898	5.041	5.122	4.060	3.412
			Avg	9.2	8.07	7.308	6.026	8.091	9.61	7.578	8.495	6.777	5.902
	GSA	CI	Mid	7.23	4.676	4.378	3.279	4.656	4.844	4.499	4.899	4.240	2.759
			Avg	10.943	7.192	7.25	5.881	7.163	8.312	6.140	6.164	6.124	5.754

strategy within 500/1000 queries to find the stepsize that minimizes the noise magnitude for each image. Since CISA adjusts the stepsize adaptively, the first adversarial example found is returned during the iterative process. Random attack also stops after the first adversarial example is found. For example, the value of the 'I-FGSM' row and 'EVO' column on the right side of Table 3 means: for the ImageNet dataset, when the target model is densenet and the substitute model is senet, if PAM, TAM, NCM and STM are embedded by binary search, I-FGSM, Evolutionary Attack

and Greedy Search, respectively, the median of the noise magnitude calculated using ℓ_2 norm on the 1000 adversarial examples is 5.568, and the mean is 8.277. In addition, if a method fails to find the adversarial example after 1000 queries on an image, we set the noise magnitude for this image to 80 as a penalty.

Tables 1 2, 3, 4, and 5 demonstrate the advantages of the new black-box adversarial attack framework and the advantages of CISA under the black-box attack setting with limited number of queries. First, for the same transfer-based

TABLE 5
Comparison Under the New Black-Box Adversarial Attack Framework on CIFAR-10 With res-101 as Target and vgg-16 as Substitute

			Target: res-101						Substitute: vgg-16				
			STM	N/A	NCM								Relax
					Greedy Search								
PAM			Vanilla	Boundary	Whey	BBA	HSJA	EVO	Tangent	Sign-OPT	QEBA	CS	
TAM	N/A	Random	Mid	2.477	0.727	1.102	0.644	1.509	0.724	1.080	1.864	1.046	0.631
			Avg	2.738	0.791	1.221	0.760	1.735	0.790	1.636	2.431	1.614	0.721
		DDN	Mid	0.434	0.405	0.370	0.354	0.424	0.406	0.364	0.365	0.354	0.338
			Avg	0.510	0.454	0.432	0.405	0.497	0.455	0.455	0.472	0.433	0.389
		FGSM	Mid	1.159	0.596	0.623	0.596	0.937	0.590	0.528	0.722	0.528	0.452
			Avg	3.157	1.249	1.636	1.414	1.866	1.248	1.756	2.906	1.775	1.134
		I-FGSM	Mid	0.426	0.393	0.361	0.337	0.416	0.394	0.335	0.341	0.327	0.325
			Avg	0.538	0.445	0.436	0.399	0.502	0.445	0.476	0.494	0.454	0.378
	Binary Search	MI-FGSM	Mid	0.469	0.413	0.394	0.357	0.460	0.419	0.430	0.439	0.400	0.341
			Avg	0.602	0.479	0.467	0.435	0.580	0.478	0.619	0.685	0.578	0.405
		vr-IGSM	Mid	0.439	0.384	0.346	0.328	0.422	0.382	0.342	0.363	0.319	0.317
			Avg	0.549	0.428	0.405	0.371	0.514	0.427	0.497	0.526	0.467	0.349
		C&W	Mid	0.576	0.389	0.372	0.325	0.418	0.386	0.462	0.401	0.460	0.319
			Avg	4.466	12.337	12.331	12.294	12.374	12.336	13.356	13.209	13.350	12.291
		EAD	Mid	0.398	0.431	0.348	0.312	0.462	0.430	0.322	0.398	0.314	0.299
			Avg	0.657	0.464	0.400	0.357	0.543	0.464	0.388	0.430	0.379	0.366
	GSA	CI	Mid	0.388	0.365	0.328	0.309	0.394	0.365	0.332	0.365	0.329	0.297
			Avg	0.468	0.408	0.395	0.357	0.464	0.408	0.415	0.464	0.399	0.341

TABLE 6
Median and Average ℓ_2 Distance of Adversarial Perturbations Between Four Models on Tiny-Imagenet

Methods	Target Sub	res-18		inc-v3			inc-res		nasnet		
		inc-res	nasnet	res-18	inc-res	nasnet	res-18	inc-v3	res-18	inc-v3	inc-res
FGSM	Mid	2.988	4.228	3.247	2.863	3.821	5.03	2.478	3.523	2.792	2.782
	Avg	5.363	5.342	4.157	4.451	5.975	6.329	3.57	4.253	4.016	4.354
I-FGSM	Mid	1.387	1.654	1.352	1.828	1.408	1.435	1.136	1.917	2.018	1.439
	Avg	2.409	2.327	2.223	1.861	2.005	1.914	1.415	2.994	2.25	2.195
MI-FGSM	Mid	1.62	2.337	1.718	1.594	2.053	1.699	1.531	2.271	2.202	1.764
	Avg	3.023	3.354	2.583	2.354	3.117	2.596	1.882	3.397	3.095	2.629
vr-IGSM	Mid	1.382	1.977	1.399	1.522	1.852	1.466	1.11	1.839	1.761	1.618
	Avg	2.825	2.719	2.273	2.156	2.347	1.938	1.569	2.997	2.765	2.381
Random	Mid	3.401			5.464		5.600			6.499	
	Avg	4.735			5.940		5.554			6.438	
CI	Mid	1.634	2.421	1.732	1.427	1.812	1.94	1.905	2.72	2.01	2.146
	Avg	3.029	3.094	2.343	2.224	2.514	2.385	2.041	3.505	2.923	3.456
Random+ Boundary	Mid	1.007			1.302		1.363			2.082	
	Avg	1.758			1.888		2.073			2.678	
CI+	Mid	0.798	0.975	1.01	0.906	0.877	1.158	1.144	1.382	1.568	1.392
	Avg	1.605	1.264	1.165	1.071	1.105	1.385	1.017	1.921	1.901	2.315
Random+ Whey	Mid	1.348			1.936		1.820			2.809	
	Avg	2.244			2.612		2.479			3.293	
CI+	Mid	0.704	0.909	0.766	0.675	0.727	0.903	0.869	1.361	1.123	1.005
	Avg	1.656	1.175	1.023	0.975	1.121	1.238	0.806	1.791	1.665	1.992
Random+ BBA	Mid	0.848			1.519		1.105			1.771	
	Avg	1.596			1.987		1.843			2.345	
CI+	Mid	0.553	0.616	0.74	0.638	0.571	0.55	0.597	0.948	1.065	0.966
	Avg	1.359	0.956	0.876	0.87	0.939	1.071	0.761	1.42	1.41	1.75
Random+ Evo	Mid	1.035			1.443		1.198			1.994	
	Avg	1.728			1.887		2.011			2.694	
CI+	Mid	0.719	0.897	0.952	0.888	0.948	1.104	1.136	1.428	1.573	1.354
	Avg	1.573	1.274	1.051	1.093	1.115	1.396	1.054	1.969	1.873	2.319
Random+ Tangent	Mid	1.634			0.936		1.054			2.025	
	Avg	3.044			1.528		2.315			2.388	
CI+	Mid	1.362	1.387	0.891	0.890	0.879	1.187	1.141	1.186	1.517	0.959
	Avg	2.393	1.865	1.493	1.424	1.340	1.849	0.925	1.469	1.751	2.099
Random+ Sign-OPT	Mid	2.270			2.131		1.316			3.343	
	Avg	3.940			1.989		2.050			3.100	
CI+	Mid	1.412	1.390	1.096	1.143	0.942	1.343	1.306	1.229	1.789	1.073
	Avg	2.791	2.170	1.736	1.729	1.575	2.198	1.183	1.692	2.015	2.340
Random+ QEBA	Mid	1.259			1.383		0.871			1.446	
	Avg	2.771			1.438		2.040			2.025	
CI+	Mid	1.230	1.185	0.774	0.833	0.808	0.958	0.907	1.095	0.987	0.916
	Avg	2.236	1.745	1.385	1.276	1.226	1.643	0.857	1.446	1.449	1.948
CISA	Mid	0.398	0.526	0.522	0.523	0.531	0.49	0.474	0.883	0.948	0.732
	Avg	1.354	0.778	0.757	0.728	0.803	0.861	0.591	1.403	1.406	1.736

attack, i.e., the same row in a table, adversarial examples generated using all the queries for binary search (Vanilla) is larger than that using half the queries for noise compression. Using the same 500 queries and the same intermediate adversarial examples, CISA's Customized Sampling generates adversarial examples with smaller noise magnitude than other decision-based attacks, even if a substitute model is not included (Random). This suggests the advantage of NCM using decision-based attack for noise compression, and also shows the higher compression efficiency of Customized Sampling with Transition Function Relaxation over existing transfer-based attacks.

Second, for the same decision-based attack, i.e., the same column of a table, final noise magnitude using Gaussian noise as the starting point for noise compression (Random) is larger than using the intermediate adversarial example

generated by transfer-based attacks in most cases. CISA can find adversarial examples with fewer query number and set aside more compression times for the subsequent noise compression. Therefore, with the same subsequent decision-based attack, the noise magnitude of 'CI' is smaller than that of other transfer-based attack methods. This verifies the advantage of PAM using transfer-based attacks to generate intermediate adversarial examples, and also illustrates the advantage of Gaussian Stepsize Adjustment and Iterative Trajectory Diversification in CISA.

5.3 Black-Box Attack on Multiple Models

To further verify the advantage of CISA over existing black-box attacks on different model structures, we report the median and average adversarial perturbation on Tiny-Imagenet and ImageNet in Tables 6 and 7, respectively. The

TABLE 7
Median and Average ℓ_2 Distance of Adversarial Perturbations Between Four Models on ImageNet

Methods	Target Sub	res-101		dense		vgg-19			senet		
		dense	senet	res-101	vgg-19	res-101	dense	senet	res-101	dense	vgg-19
FGSM	Mid	9.947	20.746	13.646	9.727	7.124	5.928	10.063	8.102	14.832	10.179
	Avg	17.305	28.495	24.193	21.197	12.276	14.77	18.162	17.664	17.609	24.432
I-FGSM	Mid	2.962	3.748	3.079	3.904	1.117	1.051	3.283	2.864	4.221	5.641
	Avg	3.753	7.655	3.513	5.424	2.067	2.494	4.401	3.788	4.614	10.593
MI-FGSM	Mid	3.242	4.128	2.996	5.643	1.271	1.144	2.936	3.238	5.056	8.013
	Avg	4.859	9.082	4.736	7.377	2.655	3.329	5.169	5.407	7.43	13.514
vr-IGSM	Mid	3.027	3.772	2.881	4.133	1.206	1.245	2.923	3.184	3.729	6.716
	Avg	4.578	9.849	3.965	6.797	2.606	2.927	5.41	5.254	6.285	13.084
Random	Mid	40.825		32.294			12.603			37.274	
	Avg	41.168		36.379			25.755			40.024	
CI	Mid	9.904	9.777	6.544	10.429	3.008	4.569	3.737	10.594	8.503	10.512
	Avg	9.512	12.658	5.989	10.821	3.737	5.341	5.768	9.733	8.582	12.497
Random+ Boundary	Mid	15.222		10.937			3.975			14.309	
	Avg	20.629		17.609			11.826			22.088	
CI+	Mid	5.557	6.06	4.693	4.283	1.643	2.06	2.51	4.545	6.387	6.872
	Avg	5.685	8.251	4.601	6.45	2.603	2.844	3.902	5.642	6.433	8.719
Random+ Whey	Mid	23.861		16.601			7.587			23.057	
	Avg	31.472		28.186			18.449			31.356	
CI+	Mid	5.776	6.656	4.733	5.789	1.761	2.409	2.808	5.179	6.332	7.773
	Avg	5.758	8.706	4.573	6.509	2.558	2.917	3.844	5.576	6.272	8.844
Random+ BBA	Mid	12.948		10.403			3.629			13.285	
	Avg	16.830		16.675			12.111			20.141	
CI+	Mid	4.394	4.624	3.651	3.047	1.275	1.57	2.15	3.788	5.645	5.192
	Avg	4.563	6.017	3.771	5.127	2.237	2.462	3.432	4.879	5.258	6.982
Random+ Evo	Mid	15.337		10.512			3.815			15.084	
	Avg	20.368		17.385			10.878			21.370	
CI+	Mid	5.491	5.74	4.774	4.726	1.852	2.175	2.631	4.088	6.328	6.439
	Avg	5.697	8.415	4.66	6.262	2.66	2.812	3.911	5.266	6.152	8.904
Random+ Tangent	Mid	15.766		14.868			3.615			12.160	
	Avg	26.493		12.048			10.698			24.628	
CI+	Mid	6.521	5.088	3.974	5.782	1.663	2.102	2.911	5.061	6.215	6.875
	Avg	6.104	8.046	4.464	6.608	2.405	3.016	3.226	5.485	6.422	8.831
Random+ Sign-OPT	Mid	18.272		17.582			3.989			12.514	
	Avg	31.470		25.284			11.014			29.122	
CI+	Mid	7.436	7.235	4.645	9.990	2.051	2.364	2.987	5.334	6.458	7.944
	Avg	7.202	8.214	5.140	9.529	2.774	3.854	3.576	6.562	7.729	10.605
Random+ QEBA	Mid	13.179		9.352			3.198			9.859	
	Avg	23.686		10.218			10.486			21.760	
CI+	Mid	5.413	3.959	4.096	4.729	1.549	1.563	2.525	4.461	5.382	5.933
	Avg	5.506	7.973	4.306	6.491	2.199	2.453	2.856	5.059	6.150	7.975
CISA	Mid	2.585	3.209	2.209	2.317	0.844	0.887	0.762	2.081	2.418	2.962
	Avg	3.646	5.821	3.421	4.193	1.763	1.717	2.645	3.743	3.746	6.013

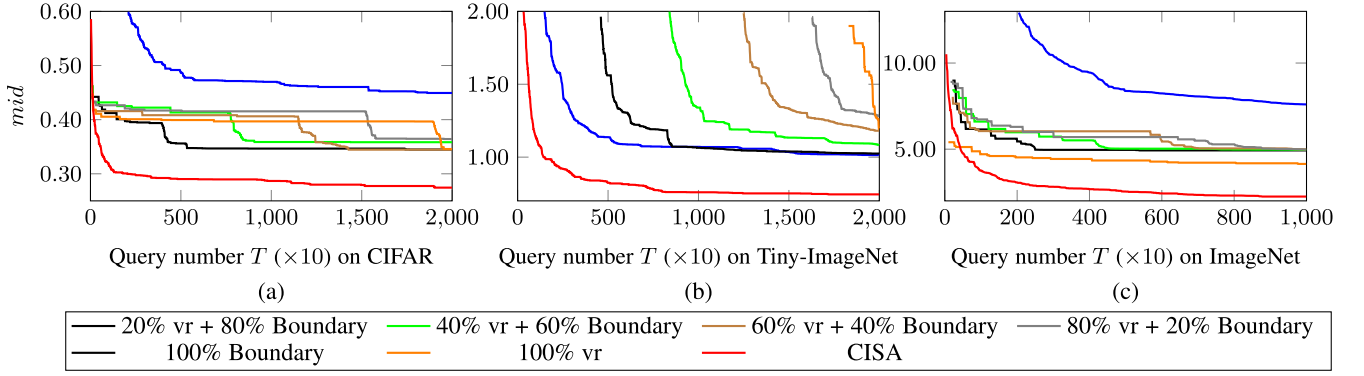
first rows of Tables 6 and 7 represent four target models with different structures. The second rows of two tables represent three substitute models with different structures for each target model. We compare the average (Avg) and median (Mid) noise magnitude generated by CISA and other 14 attacks on different target-substitute model combinations.

It can be seen from Tables 6 and 7 that CISA achieves the smallest median and average noise magnitude in ℓ_2 norm on different target models with the same number of queries than all the other methods on both datasets. After replacing random Gaussian noises with the adversarial examples generated by CISA as the starting point for noise compression, the noise magnitudes of all decision-based attacks drastically reduce. This confirms the conclusion on combining

transfer-based and decision-based attacks in Section 3.3. Meanwhile, compared with original Boundary Attack (Random+Boundary), CISA has a significant decrease on noise magnitude, which validates the effectiveness of customizing the distribution with current noise and failed samplings.

5.4 Ablation Study

The number of queries T is the most important parameter in the black-box adversarial attack with limited query. We compare the noise compression ability of CISA and other decision-based attacks under more queries. First, we use the PAM and TAM of CISA (CI) to generate intermediate adversarial examples on 10000 images on the ImageNet dataset, and use different decision-based attacks to conduct 10000 queries on each adversarial example. The substitute model


 Fig. 8. Median ℓ_2 distance of adversarial noise under different query number T .

of CI is densenet-161, and the target model is vgg-19. We average the noise ratio of 10000 images relative to the initial noise generated by CI at each query

$$\text{Mean noise ratio} = \frac{1}{n_{\text{data}}} \sum_{i=1}^{n_{\text{data}}} \left(\left\{ \frac{\|x'_i - x_i\|_2}{\|x^*_i - x_i\|_2} \mid x \in \mathbf{X} \right\} \right), \quad (25)$$

where x^* is the intermediate adversarial example generated by CI. Curves of mean noise ratio as T increases is shown in Fig. 9. A higher T provides decision-based attacks with more opportunities to fine-tune the adversarial noise. It can be seen that mean noise ratio of CISA is lower than other decision-based attacks under different query times. CISA requires about 2000 queries to achieve a mean noise ratio of 20%, while other decision-based attacks require 10000 queries or more, which demonstrates the efficiency in noise compression of Customized Sampling.

We combine vr-IGSM and Boundary which achieve small noise magnitude from transfer-based attacks and decision-based attacks, and compare the query efficiency with CISA under different query number allocation ratios in Fig. 8. We compare the median noise magnitude of all adversarial examples on CIFAR-10, Tiny-Imagenet and Imagenet. The target and substitute models are vgg-16 and resnet-101 for CIFAR-10, inception-v3 and resnet-18 for Tiny-Imagenet, and resnet-101 and vgg-19 for ImageNet. The total query numbers of one image for CIFAR-10, Tiny-Imagenet and ImageNet are 20000, 20000, and 10000, respectively. We compare 6 different query number allocation ratios. For example, the black curves in Fig. 8 of '20% vr + 80% Boundary' means that 20% of the total query number are used for vr-IGSM, and the remaining 80% are used for noise

compression with Boundary attack based on the generated intermediate adversarial examples. It can be seen that CISA represented by the red curve find adversarial examples with a smaller noise magnitude than all the combinations of vr-IGSM and Boundary on different query numbers and datasets.

There is a noticeable difference in noise compression between transfer-based attacks and decision-based attacks in Fig. 8. Initially, vr-IGSM can quickly generate adversarial examples with small noise magnitude. However, as the binary search progresses, the efficiency of noise compression decreases, and diminishing marginal effect appears. When the query number of vr-IGSM is used up and Boundary Attack begins, the noise compression efficiency will increase sharply (obvious in Fig. 8a). When the noise is compressed to a certain degree, the number of adversarial examples with smaller noise drops, and the query efficiency slows down again. Therefore, a reasonable strategy is to use transfer-based attack in the initial stage to find adversarial examples with small noise magnitude at the cost of small number of queries, and then use other queries for noise compression. Since CISA's PAM uses Gaussian Stepsize Adjustment to dynamically allocate the query number of TAM for each image, its query efficiency is significantly improved compared with the combination of vr-IGSM and Boundary with fixed query number allocation ratio.

In order to further verify the effectiveness of each module in CISA, we conduct ablation study on Tiny-Imagenet with 4 substitute-target model combinations. The median and average noise magnitudes without each module of CISA are reported in Table 8. For NCM, we additionally subdivide Customized Sampling into three steps of variance customization, mean customization, and stepsize customization. It can be seen that all the steps in the four modules contribute to a smaller noise magnitude. Simply removing any of the steps may lead to larger adversarial noise.

Adversarial training [58] and model ensemble are two widely used defend methods. In Table 10, we use resnet18 as substitute models to attack 3 adversarially trained models (densenet-161, inception v4 [53] and vgg-19) and ensemble model consisting of three models on Tiny-Imagenet. Although defence methods increase the difficulty of adversarial attack, the noise magnitude of adversarial examples generated by CISA is still much lower than other attacks. We report the success rates of 9 transfer-based attacks on Tiny-ImageNet in Table 9. Subsequent decision-based attacks do

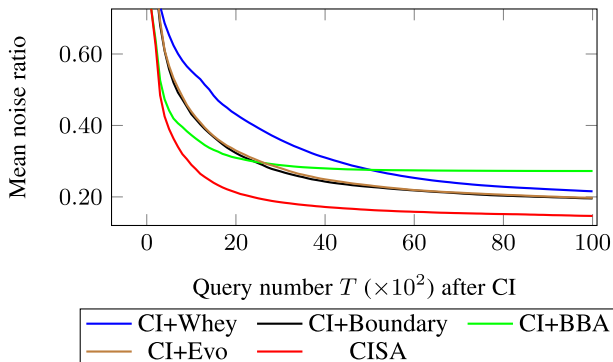

 Fig. 9. Mean noise ratio under different query number T .

TABLE 8
Ablation Study of CISA on Tiny-ImageNet

Target Model	inc-res		inc-v3		nasnet		inc-res	
Substitute Model	res-18		nasnet		inc-res		inc-v3	
Evaluation Criterion	median	average	median	average	median	average	median	average
CISA w/o Gaussian Stepsize Adjustment	0.824	1.232	1.048	1.59	1.054	1.876	0.811	1.229
CISA w/o Iterative Trajectory Diversification	0.914	1.3	1.121	1.705	1.152	2.033	0.864	1.238
CISA w/o Variance Customization	1.151	1.692	1.206	1.527	1.597	2.479	1.079	1.288
CISA w/o Mean Customization	0.834	1.176	1.061	1.465	1.054	1.813	0.853	1.197
CISA w/o Stepsize Customization	0.722	1.099	0.906	1.535	0.923	1.819	0.57	0.79
CISA w/o Transition Function Relaxation	0.888	1.299	1.095	1.794	1.146	2.081	0.856	1.252
CISA	0.49	0.861	0.531	0.803	0.732	1.736	0.474	0.591

TABLE 9
Success Rate of 9 Transfer-Based Attacks on Tiny-ImageNet

Target Sub	res-18			inc-v3		
	inc-v3	inc-res	nasnet	res-18	inc-res	nasnet
Random	99.6%	99.2%	99.9%	99.6%	99.3%	99.4%
DDN	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%
FGSM	98.5%	97.4%	99.0%	98.8%	98.6%	99.1%
I-FGSM	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%
MI-FGSM	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%
vr-IGSM	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%
C&W	97.3%	98.3%	77.2%	97.5%	98.3%	70.9%
EAD	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%
CI	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%

not influence the success rate. Since the goal of transfer-based attacks in this paper is to generate intermediate adversarial examples for further noise compression, the success rate of 100% for almost all methods can no longer effectively measure the performance of different attacks. Therefore, we take

minimizing the noise magnitude as the goal of black-box attack to reduce the perceptibility of adversarial examples for humans and adversarial defense methods.

Fig. 10 compares adversarial examples generated by five different black-box adversarial attacks on ImageNet. The first image of each row is the original image. The second and third images of each row are the intermediate adversarial examples and their noises. The ‘Noise (CISA)’ and ‘CISA’ columns are the noises and final adversarial examples of CISA after Customized Sampling. The next three columns show adversarial examples of I-FGSM, Boundary and BBA. The last column compares the noise magnitude of five adversarial attacks. CISA performs a total of 1000 queries on the target model on each image. All other attacks also perform 1000 queries on the target model for each adversarial example. Resnet-101 and vgg-19 are used as target model and substitute model for I-FGSM and CI, respectively. Compared with I-FGSM, Boundary and BBA, CISA generates adversarial examples with much smaller noise magnitude.

TABLE 10
Median and Average ℓ_2 Distance of Adversarial Perturbation Against Adversarially Trained Models and Ensemble Model on Tiny-Imagenet

Methods	Target sub	dense (adv)	inc-v4 (adv)	vgg-19 (adv) res-18	inc-v3+ inc-res v2+ nasnet
I-FGSM	Mid	3.984	5.646	5.835	4.206
	Avg	16.39	21.081	22.346	15.479
Random+	Mid	2.534	3.033	2.992	3.215
Boundary	Avg	3.832	4.195	4.229	4.49
Random+	Mid	3.615	4.13	4.017	4.251
Whey	Avg	4.75	5.336	5.314	5.202
Random+	Mid	2.069	2.452	2.315	3.014
BBA	Avg	3.059	3.507	3.538	3.971
Random+	Mid	2.478	2.93	2.938	3.204
Evo	Avg	3.864	4.208	4.228	4.353
Random+	Mid	2.495	3.356	3.245	3.411
Tangent	Avg	3.230	4.517	3.408	4.457
Random+	Mid	3.268	3.650	3.600	3.839
Sign-OPT	Avg	4.197	4.932	4.689	4.589
Random+	Mid	2.217	3.036	2.391	2.902
QEBA	Avg	3.097	4.221	3.114	4.389
Random+	Mid	1.678	1.93	1.981	1.621
CS	Avg	2.293	2.479	2.431	2.182
CISA	Mid	1.261	1.446	1.393	1.254
	Avg	2.093	2.195	2.09	2.01

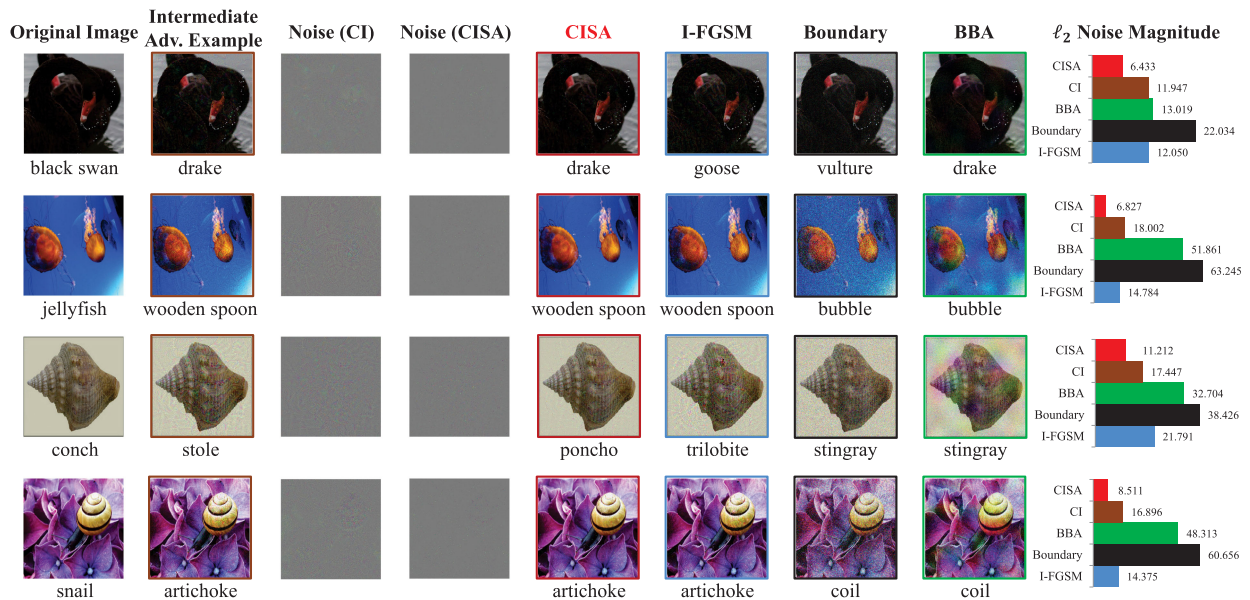


Fig. 10. Comparison of adversarial examples generated by CISA, I-FGSM, Boundary, and BBA on ImageNet dataset. The intermediate and final adversarial noises are demonstrated as 'Noise (CI)' and 'Noise (CISA),' respectively. The labels and misclassification categories are noted under original images and adversarial examples. The rightmost column compares the noise magnitude of five adversarial attacks in ℓ_2 norm.

6 CONCLUSION

In this paper, we propose a new framework consisting of four modules for black-box adversarial attack with limited queries. Based on the new framework, we propose CISA for query-efficient black-box adversarial attack. CISA uses Gaussian Stepsize Adjustment in Parameter Adjustment Module to adaptively set stepsize for transfer-based attack, and diversifies the iterative trajectory by combining gradient ascent and descent in the Transfer Attack Module. The intermediate adversarial examples generated by Transfer Attack Module are used as the starting point for subsequent refinement under a small number of queries. In Noise Compression Module, CISA adjusts the variance and mean of sampling distribution according to current adversarial noise, and customizes the stepsize as well as mask to achieve efficient noise compression. Based on the theoretical analysis of monotonicity in noise compression, we further relax the transition function of CISA to accelerate noise compression in State Transition Module. Experiments on three image classification datasets verify the feasibility and generalizability of the new black-box adversarial attack framework to improve the query efficiency. The experimental results also demonstrate that CISA generates adversarial examples with much smaller noise magnitude in ℓ_2 norm than existing transfer-based and decision-based attacks.

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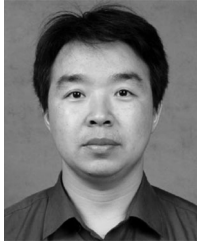
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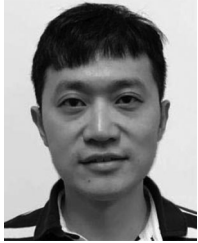
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